📘 **AI/ML Internship – Day 6 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: Problem Solving Approach (Very Important)**

🔹 **How to Solve Programming Problems?**

👉 Follow these steps:

1. **Understand the problem**
2. **Break into smaller steps**
3. **Write logic (pseudo code)**
4. **Convert to Python code**
5. **Test with inputs**

🔹 **Example Problem**

👉 Find sum of numbers from 1 to N

Step 1: Logic

- Take input N
- Loop from 1 to N
- Add numbers

Step 2: Code

```
n = int(input("Enter number: "))
total = 0

for i in range(1, n+1):
    total += i

print("Sum:", total)
```

📌 **Part 2: Working with Lists (Advanced Logic)**

🔹 **Find Maximum Number**

```
nums = [10, 20, 5, 40]

max_num = nums[0]

for n in nums:
    if n > max_num:
        max_num = n

print(max_num)
```

🔹 **Find Second Largest Number**

```
nums = [10, 20, 5, 40]

nums.sort()
print(nums[-2])
```

🔹 **Remove Duplicates**

```
nums = [1, 2, 2, 3, 4, 4]

unique = list(set(nums))
print(unique)
```

📌 **Part 3: String Problem Solving**

🔹 **Reverse a String**

```
text = "python"
print(text[::-1])
```

🔹 **Count Characters**

```
text = "hello"

count = {}

for ch in text:
    count[ch] = count.get(ch, 0) + 1
```

```
print(count)
```

🔹 **Check Palindrome**

```
text = input("Enter string: ")

if text == text[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

📌 **Part 4: Working with Dictionaries (Advanced)**

🔹 **Find Top Student**

```
students = [
    {"name": "A", "marks": 70},
    {"name": "B", "marks": 90}
]

top = students[0]

for s in students:
    if s["marks"] > top["marks"]:
        top = s

print(top)
```

🔹 **Group Data**

```
marks = [50, 80, 90, 40]

passed = [m for m in marks if m >= 50]
print(passed)
```

📌 **Part 5: Menu-Driven Programs (Very Important)**

🔹 **What is Menu Driven Program?**

A program that shows options and performs actions based on user choice.

🔹 **Example**

```
while True:
    print("1. Add")
    print("2. Exit")

    choice = int(input("Enter choice: "))

    if choice == 1:
        a = int(input())
        b = int(input())
        print("Sum:", a+b)

    elif choice == 2:
        break
```

📌 **Part 6: Real-World Thinking**

👉 In AI:

- Data comes as lists/dictionaries
- We process, filter, analyze

👉 Today's focus:
✓ Logic building
✓ Data processing
✓ Problem solving

🎯 **Tasks for Day 6**

🧠 **Task 1: Theory**

1. Steps in problem solving
2. What is algorithm?
3. What is menu-driven program?

4. Why logic is important in AI?
5. Difference between list and dictionary

💻 **Task 2: List Problems**

1. Find maximum and minimum
2. Find second largest
3. Remove duplicates
4. Count even and odd numbers

💻 **Task 3: String Problems**

1. Reverse string
2. Check palindrome
3. Count characters
4. Count vowels and consonants

💻 **Task 4: Dictionary Problems**

1. Create student list
2. Find topper
3. Calculate average
4. Filter passed students

🔥 **Task 7: Logic Challenge**

1. Find second smallest number
2. Merge two lists without duplicates
3. Find frequency of elements